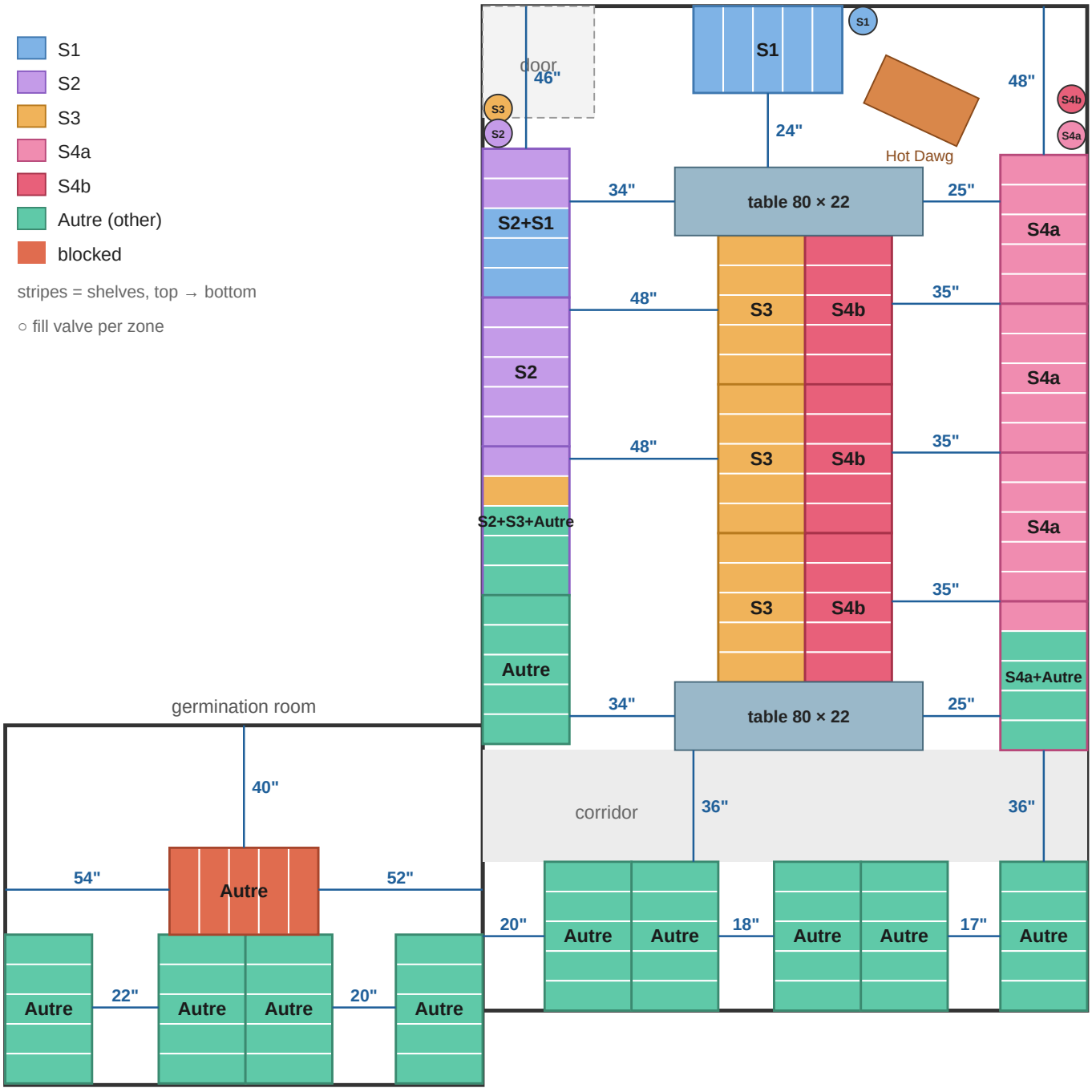
Nursery room plan



Sample · 1 rack

Flood rack

Flood rack sample — parts to find (1 rack, 5 trays)

Indoor Farmer

☐		Grow Rack Shallow Rack Tray 45,5 × 25,5 × 4,5 po	**5**
☐		Active Aqua fill/drain fitting ¾	**10**
☐		Active Aqua screen fitting	**10**
☐		Active Aqua extension fitting	**10**
☐		Active Aqua vinyl ¾ ID, 100 pi roll	**~5 m**

U.S. Solid

☐		U.S. Solid motorized ball valve ¾ in SS, 3 indicator lights, 9–36 V, 2-wire auto-return, NO	**5**

Dubois

☐		Adaptateur mâle ¾ MPT × ¾ insert	**20**
☐		Valve barbée 0,18 po pour micro-tube	**5**
☐		Coude ¼ barbé	**5**

RONA

☐		IPEX tuyau ABS 1½ po × 12 pi	**1**
☐		IPEX té sanitaire ABS 1½ po F×F×F	**10**
☐		IPEX réduction ABS 1½ × ¾ po mâle × FPT	**10**
☐		IPEX coude 90 extra-long ABS 1½ po F×F	**3**
☐		Aqua-Dynamic manchon flexible PVC 1½ × 1½ po, colliers inox	**2**
☐		IPEX té sanitaire ABS 3 × 3 × 1½ po	**1**
☐		Oatey ciment ABS jaune 473 mL	**1**

Home Depot

☐		Commercial Electric attaches zip UV 8 in, 100-pack	**~10**

Amazon

☐		Collier inox 304 19–29 mm, 20-pack	**30 clamps**
☐		HydroMaxx flexible PVC 1½ in Sch 40, white, 25 ft	**~2 ft**

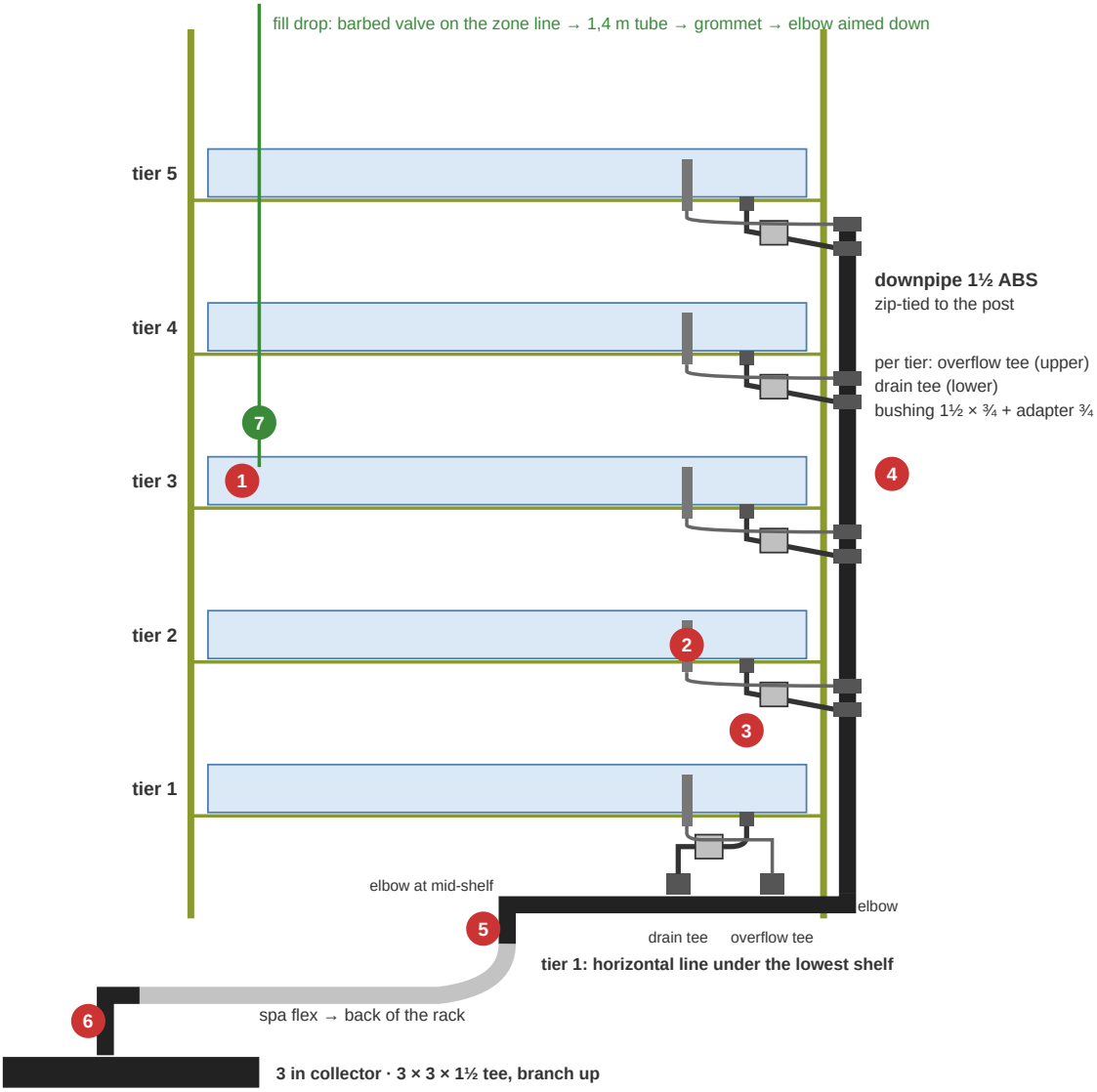
☐		Néoprène 1/16 × 2 in × 10 ft	**~10 pads**
☐		Vefhby neoprene strip 1/8 × 3 in × 10 ft, solid	**as needed**
☐		MIXC drip tubing ¼ in OD, 200 ft	**7 m**
☐		HORTIPOTS grommet ¼ po, 25-pack	**5**
☐		WAGO 221, 90-pack	**~10**

Not on hand

☐		Wire 18/2 (Orisha, not ordered)	**~8 m**

Sample — flood rack + downpipe

Spec: nursery/rack-water.md · tray-ports.md · tray-fill-drop.md · numbers = steps on the next pages



Flood rack + downpipe — parts, in order

1 · Tray + shims



Flood tray 45,5 × 25,5

→



Neoprene 1/16 in pads

→



Neoprene 1/8 in pads

Lift the flats → set the tray → pour 5 L → shim the high corners until all runs to the drain port → flats back.

2 · Ports



Fill/drain fitting ¾

→



Screen fitting (inside)



Fill/drain fitting ¾

→



Extension

→



Extension

→



Screen on top

Row 1 = drain port, flush with the floor. Row 2 = overflow standpipe ≈ 2,5 in.

3 · Drain drop + overflow drop



Drain port

→



Vinyl ¾ ID

→



Inox clamp

→



Adapter ¾ MPT × insert

→



Motorized valve ¾

→



Adapter ¾ MPT × insert

→



Vinyl ¾ ID

→



Adapter in the drain tee



Overflow port

→



Vinyl ¾ ID

→



Inox clamp

→



Adapter in the overflow tee

Téflon on every thread. One inox clamp per barb.

4 · Downpipe, per tray tier



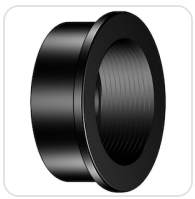
ABS 1½ pipe

→



Overflow tee 1½ (upper)

→



Bushing 1½ × ¾ SPG × FPT

→

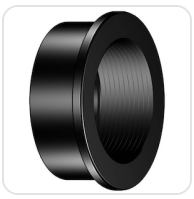


Adapter ¾ MPT × insert



Drain tee 1½ (lower)

→



Bushing 1½ × ¾ SPG × FPT

→



Adapter ¾ MPT × insert

→



Zip ties → rear post

→



ABS cement

Glue the tees per tray tier, branches toward the tray; bushings glued in the branches; zip-tie the downpipe to the rear post.

5 · Tier 1 · horizontal line under the lowest shelf



Downpipe foot → Long 90 1½ → Stub → Overflow tee 1½ → Stub →



Drain tee 1½ (branch up) → Stub to mid-shelf → Long 90 1½

Tier-1 drain + overflow drops land on these 2 tees.

6 · To the collector



Long 90 at mid-shelf → Stub → Flexible coupling → Spa flex 1½ → back of the rack → Flexible coupling →



Stub → Long 90 1½ → Stub → Collector tee 3 × 3 × 1½, branch up → 3 in collector

7 · Fill drop + valve wiring



Zone line ¾ PE

→



Barbed valve (punched in)

→



Tube ¼ in, 1,4 m

→



Grommet (3/8 in hole)

→



Barbed elbow, aimed down

Wire 18/2 from the contactor

→



WAGO 221 splice

→



Motorized valve, in parallel

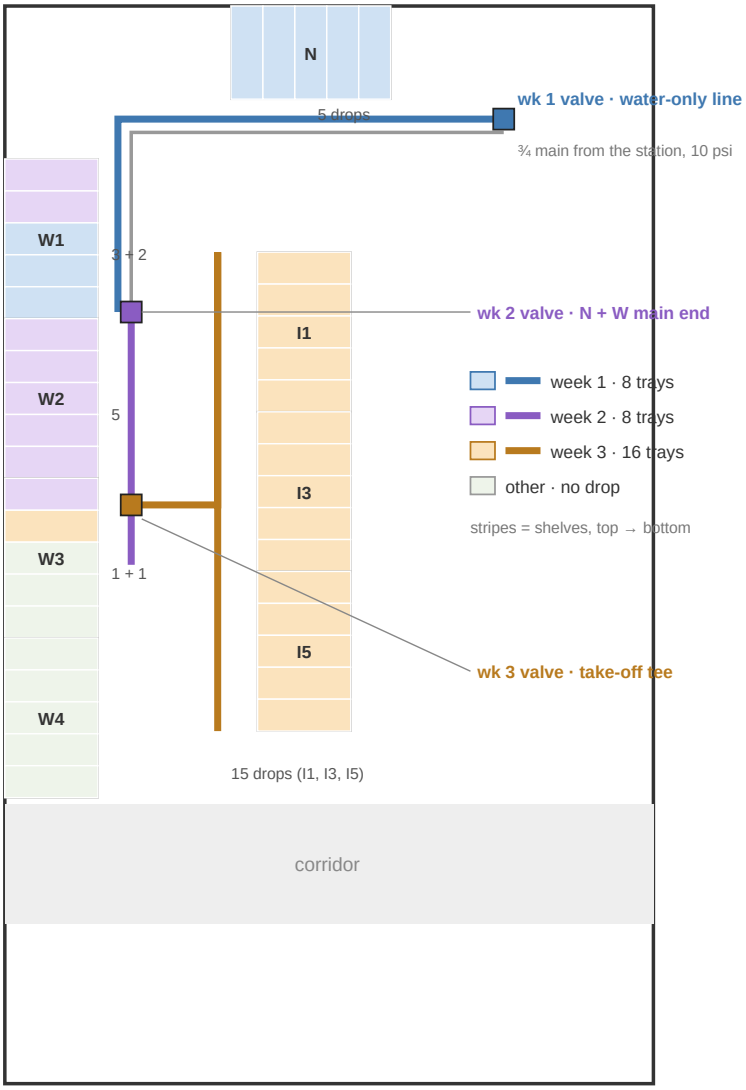
Hot-water dip to seat the tube on the barbs. Zip-tie the tube to the shelf.

Weeks 1 · 2 · 3

Fill lines

Plan — week 1 · 2 · 3 nursery fill lines

Spec: nursery/water-supply.md § Zone lines · shelf weeks from the default room plan · line routes schematic



Fill lines — parts, in order

1 · Week 1 · water-only line



Tee in the bib PE

→



Adapter ¾ MPT × insert

→



PSR-2 10 psi

→



Adapter ¾ MPT × insert

→



PE ¾, 15 ft

→



Reducing adapter 1 MPT × ¾ insert

→

Orisha fill valve 1 in (week 1)

Own 10 psi regulator: plain water, no fertigation.

2 · Each zone line (weeks 1, 2, 3)



Reducing adapter 1 MPT × ¾ insert

→

Orisha fill valve 1 in

→



Reducing adapter 1 MPT × ¾ insert

→



PE ¾ along the racks

→



Adapter ¾ MPT × insert (line end)

→



Cap ¾ FPT, unscrew to flush

Week 2 valve on the north + west main end · week 3 valve on a take-off tee.

3 · Turns + take-offs



Elbow 90 insert ¾

→



Tee insert ¾

→



Inox clamp, 1 per barb

Téflon on the threads.

4 · One drop per tray



Zone line ¾ PE

→



Barbed valve, punched 3 mm

→



Tube ¼ in, 1,4 m

→



Grommet, 3/8 in hole

→



Barbed elbow, aimed down

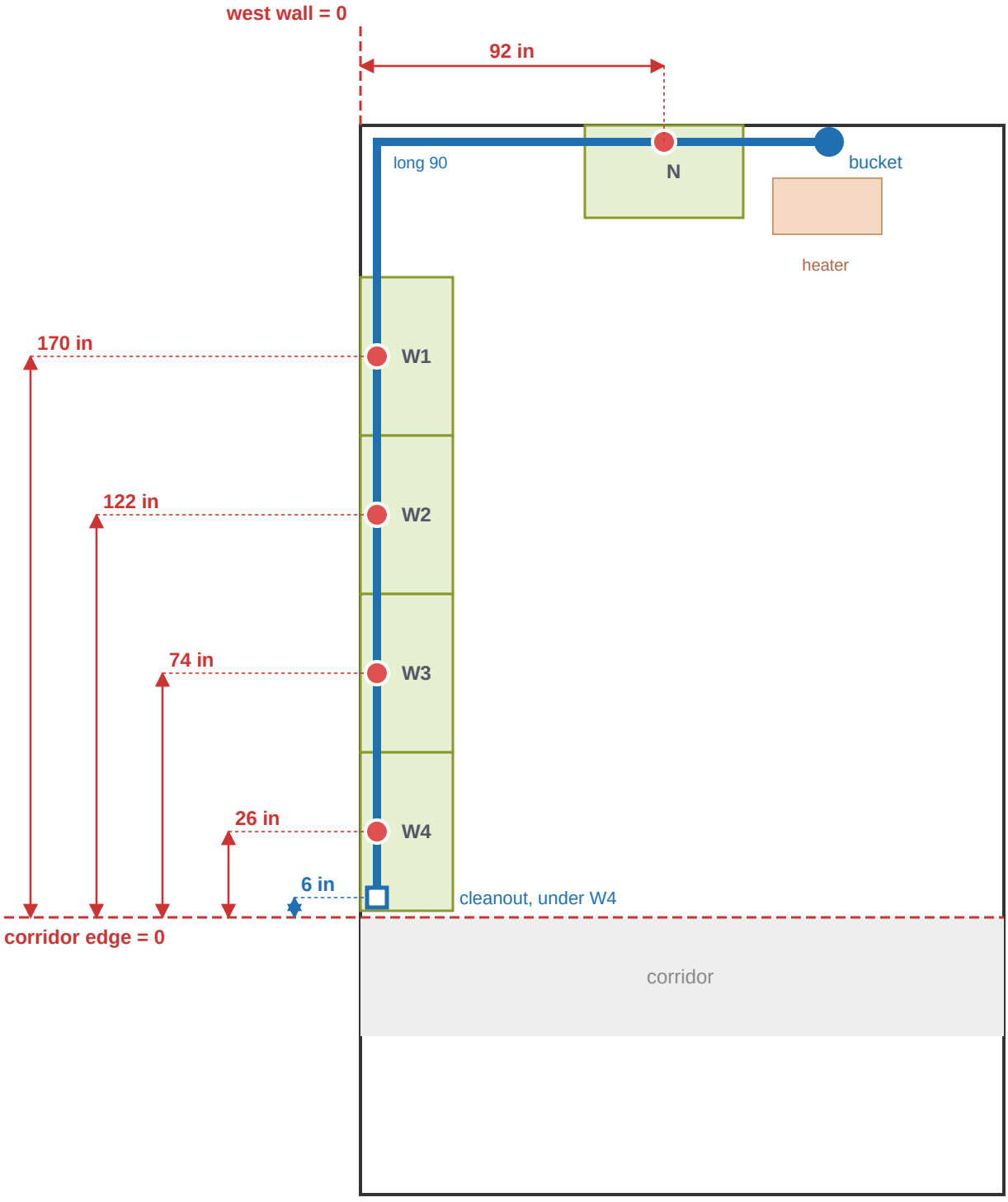
Week 1: 8 drops · week 2: 8 · week 3: 16.

West + north racks

Collector tees

Collector tees — west + north racks

Tee = centre of each rack, under its back edge (wall side). Distances to the tee centre.



Electricals

Rack wiring

Rack wiring sample — parts to find (1 rack)

Home Depot

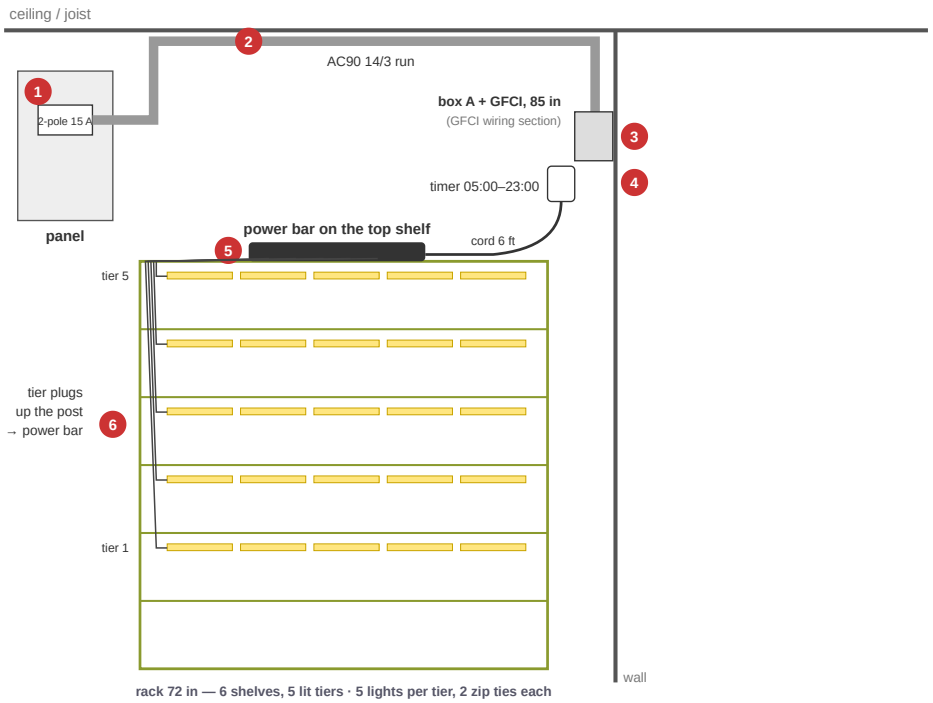
☐		Commercial Electric attaches zip UV 8 in, 100-pack	**50**

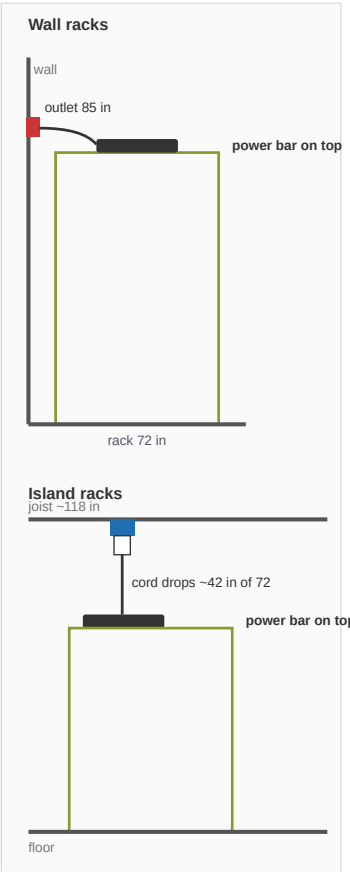
Amazon

☐		BN-LINK 7-day digital timer, 15 A	**1**
☐		DEWENWILS power bar 6 outlets, 6 ft, 14/3 SJT, 15 A, cUL, 2-pack	**1**
☐		Barrina T8 42 W 4 ft 5000 K	**25 lights**

Rack wiring

Spec: nursery/rack-lighting.md





Rack wiring — parts, in order

1 · Outlet → lights



Timer into the GFCI, 05:00–23:00

→



Power bar on the top shelf

→



T8 lights, 5 per tier

→



Zip ties, 2 per light

Tier cords up the rear post → power bar. 1 050 W ≈ 8,8 A per rack.

Electricals

GFCI wiring

GFCI wiring — parts to find (box A + box B)

RONA

☐		Eaton prise DDFT inviolable 15 A blanc, 3-pack	**2**
☐		Eaton plaque Decora 1 élément blanc, 10-pack	**2**
☐		Iberville manchons anti-court-circuit, 30	**2 (jumper ends)**

Home Depot

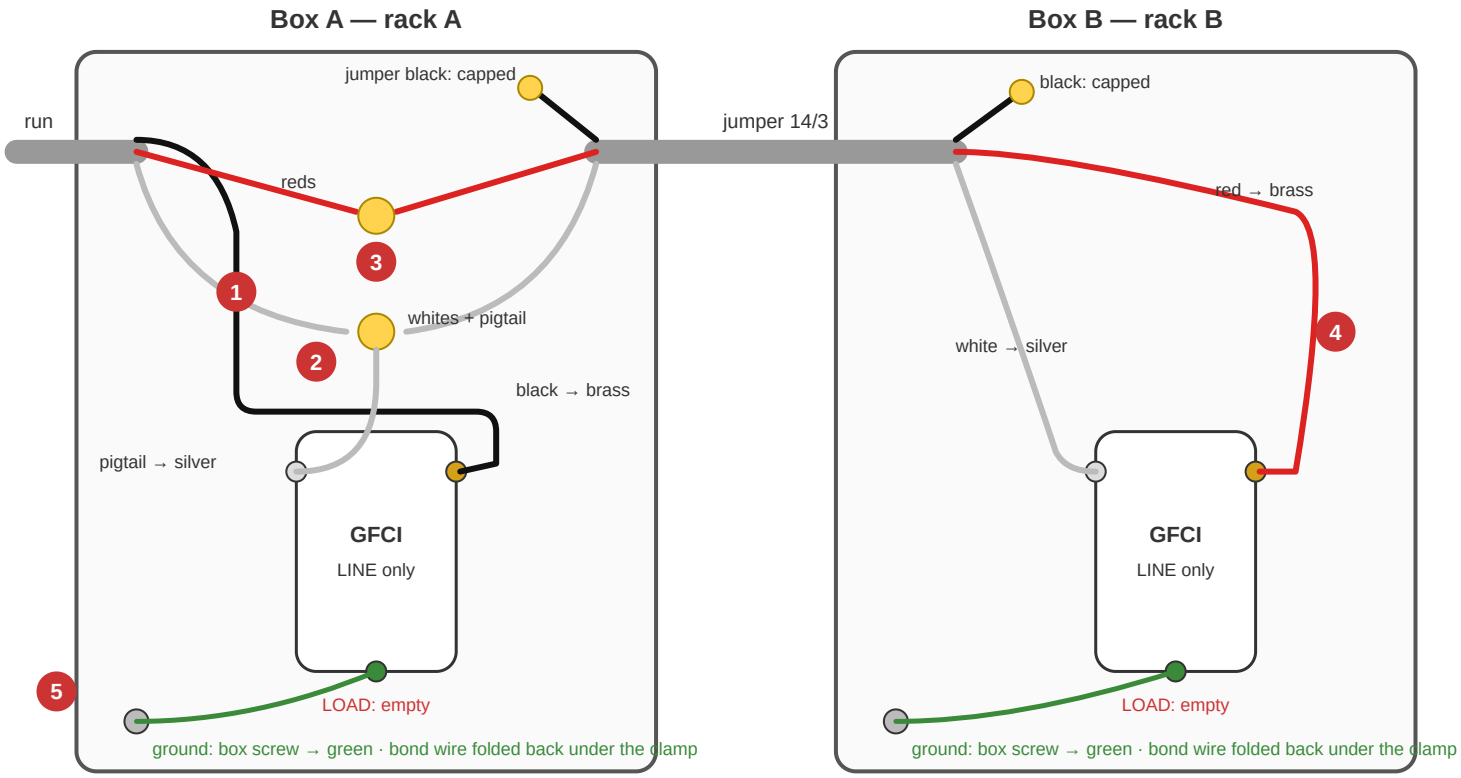
☐		Southwire AC90 14/3, 150 m	**jumper ~1,5 m**

Not on hand

☐		Wire nuts	**5**
☐		Pigtail wire 14 AWG, white + green	**~1 m each**

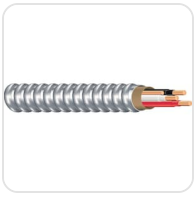
GFCI wiring — box A + box B

Spec: nursery/rack-lighting.md § Electrical · 1 run = 2 racks



GFCI wiring — connections, in order

1 · Cable into the box



AC90 14/3

→



Anti-short bushing

→



Box clamp: bond wire folded back under it

Bond wire never on a terminal.

2 · Box A



Run black

→



GFCI LINE brass

Run white + jumper white + pigtail

→



Wire nut

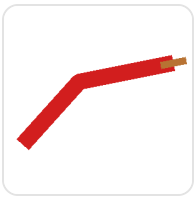
→

Pigtail

→

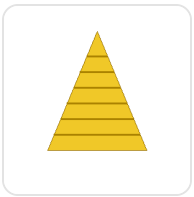


GFCI LINE silver



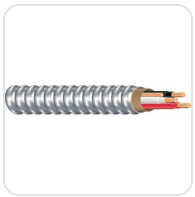
Run red + jumper red

→



Wire nut

→

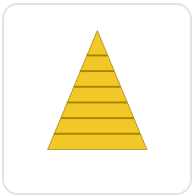


Jumper → box B



Jumper black

→



Capped

3 · Box B


Jumper red

→


GFCI LINE brass

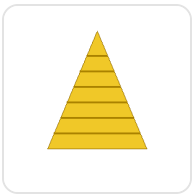
Jumper white

→


GFCI LINE silver


Jumper black

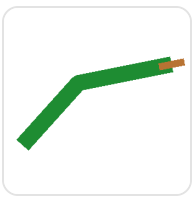
→


Capped

4 · Ground, both boxes


Box ground screw

→


Green pigtail

→


GFCI green screw

LOAD terminals stay empty.

Electricals

Outlets + runs

Outlets + runs — parts to find (10 runs, 20 boxes)

RONA

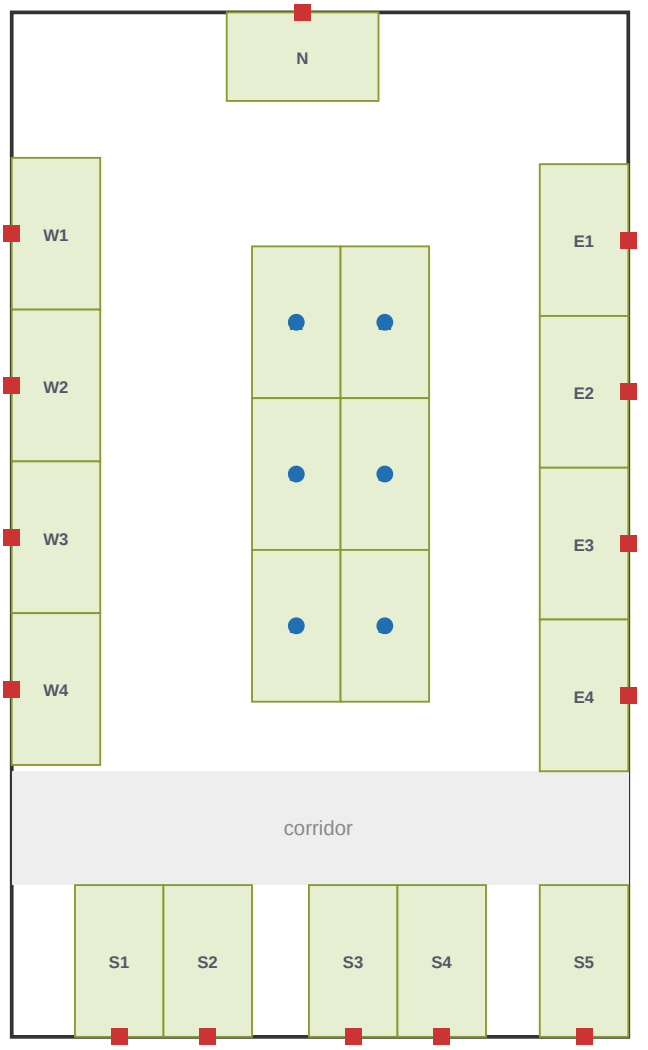
☐		Cutler-Hammer disjoncteur BR 2 pôles 15 A	**10**
☐		Iberville manchons anti-court-circuit, 30	**20 (run ends)**
☐		Iberville attache-câble 2 trous alu, 25	**~220**
☐		Precision vis à bois #8 × 1¼ po zinguées, 100	**40**

Home Depot

☐		Southwire AC90 14/3, 150 m	**2 rolls**
☐		Iberville AC90 connector ⅜ in, bag of 5	**10**
☐		Iberville boîte 1 prise 3 × 2 × 3 in, métal	**20**

Plan — outlets

Spec: nursery/rack-lighting.md · rack spots from the default room plan

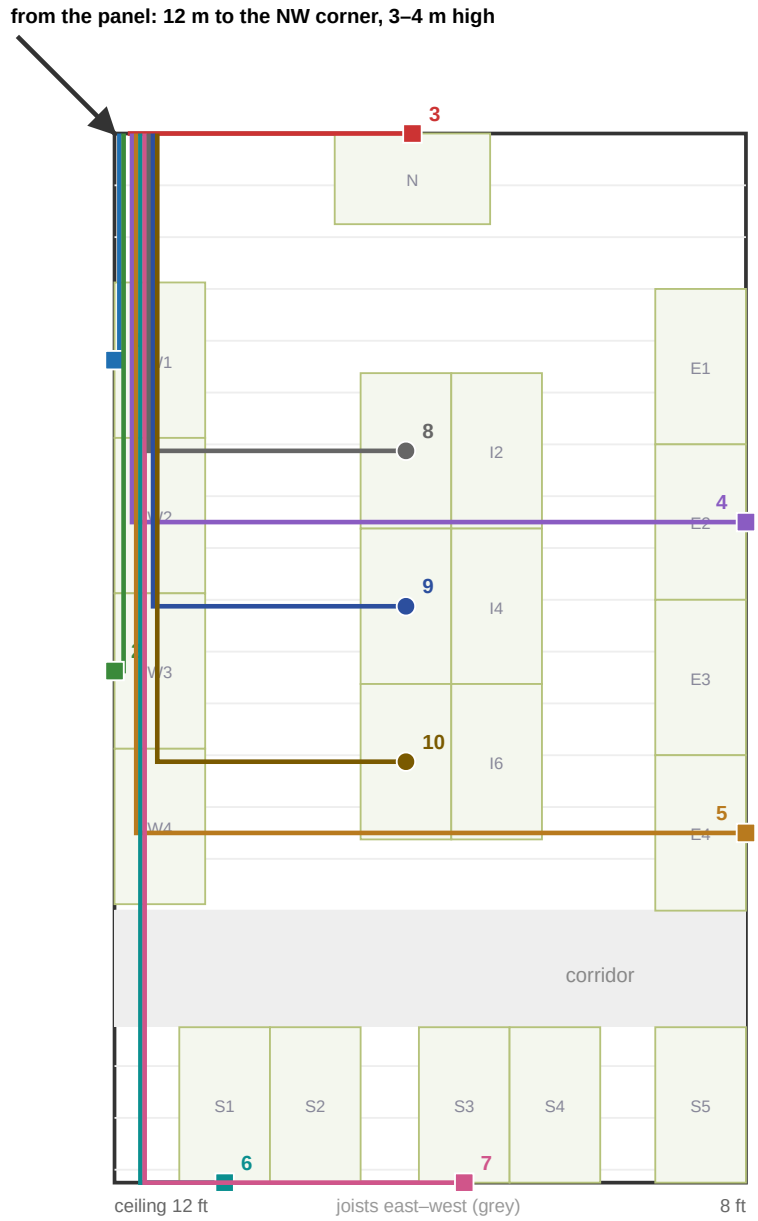


■ wall outlet at 85 in, behind the rack ● island: box on the joist above the rack
ceiling 12 ft west → 8 ft east · joists east–west

Rack	X, in	Y, in	Mount · height
Island			
I1	90	98	joist · 122 in
I2	118	98	joist · 115 in
I3	90	146	joist · 122 in
I4	118	146	joist · 115 in
I5	90	194	joist · 122 in
I6	118	194	joist · 115 in
West wall			
W1	0	70	wall · 85 in
W2	0	118	wall · 85 in
W3	0	166	wall · 85 in
W4	0	214	wall · 85 in
North wall			
N	92	0	wall · 85 in
East wall			
E1	195	72	wall · 85 in
E2	195	120	wall · 85 in
E3	195	168	wall · 85 in
E4	195	216	wall · 85 in
South wall			
S1	34	324	wall · 85 in
S2	62	324	wall · 85 in
S3	108	324	wall · 85 in
S4	136	324	wall · 85 in
S5	181	324	wall · 85 in

X = east from the west wall · Y = south from the north wall · measured from the north-west corner, to the box centre

Map — rack power runs, 10 cuts



Run	Racks	Cut
7	S3 + S4	**27 m**
5	E4 + S5	**26 m**
6	S1 + S2	**26 m**
4	E2 + E3	**24 m**
10	I5 + I6	**23 m**
9	I3 + I4	**22 m**
2	W3 + W4	**21 m**
8	I1 + I2	**20 m**
1	W1 + W2	**19 m**
3	N + E1	**19 m**
	Total	**227 m**

1. **Validate the length of the longest first by measuring it with a tape**

2. **AC90 14/3. Make sure there are 4 wires in the cable.**

3. **Start cutting the longest then work your way to the shortest run**

Outlets + runs — parts, in order

1 · Panel → box



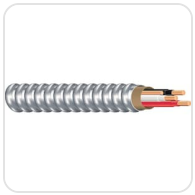
2-pole 15 A breaker, common trip

→



AC90 connector at the panel

→



AC90 14/3 run

→



Anti-short bushing, each cut end

→



Cable straps, every 1,5 m

→



Box: wall 85 in or joist, run clamped + coiled

Cut lengths: previous page.